MATH 161A - Applied Probability & Statistics

Connor Petri

Fall 2025

Contents

1. Skipped

2. Probability

2.1 Sample Spaces and Events

Probability

- Probability is used as a mathematical tool to understand or describe random phenomena, chance variation, and uncertainty.
- We will use probability as a tool to evaluate the reliability of our conclusions about the population when we only have sample information.
- Probablility is used in a variety of fields including genetics, quality control, and finance.
- Probability is used as a model for situation for which outcomes occur randomly.
- Such situations are called **experiments**.
- Examples:
  - Toss a coin.
  - Roll a die.
  - Draw a card from a deck.

Vocabulary

- An action or process whose outcome is subject to uncertainty is called an **experiment**.
- The possible outcomes of an experiment are called **sample space**.
- An **event** is a specific outcome or a set of outcomes from an experiment.
- A **simple event** is an event that consists of a single outcome from sample space S.

Example What is the outcome of rolling an even (E), rolling greater than 3 (F), and rolling a 2 (G) on a 6-sided die?

$$S = 1, 2, 3, 4, 5, 6$$
$$E = 2, 4, 6$$
$$F = 4, 5, 6$$
$$G = 2$$

A Note on Venn Diagrams All Venn Diagrams in this course require a rectangle to represent the sample space. Events are represented by circles within the rectangle.

Operations

Set Operations

- $A \subseteq B$: A is a subset of B.
- $A \cap B$: the intersection of A and B (elements in both A and B).
- $A \cup B$: the union of A and B (elements in either A or B).
- A^c or A': the complement of A (elements in S that are not in A).

Logical Operations

- A or B: $A \vee B$
- A xor B: $A \oplus B$
- A and B: $A \wedge B$
- not A: $\neg A$

Mutually Exclusive Events

- Events A and B are **mutually exclusive** if they cannot both occur at the same time.
- In terms of set operations: $A \cap B = \emptyset$.
- Example: Rolling a die and getting an even number (E) or an odd number (O) are mutually exclusive events.

Identities

$$A \cup A' = S$$
$$A \cap A' = \emptyset$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$
$$\neg(A \vee B) = \neg A \wedge \neg B$$
$$\neg(A \wedge B) = \neg A \vee \neg B$$

2.2 Probability

Definition Probability is a function that maps the sample space S to $\mathbb{R}$. The sample space is the **domain** and the real numbers are the **codomain**.

Axioms of Probability

1. Non-negativity: For any event A, $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B, $P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i)$.

$$P(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i)$$

Not axioms but useful observations

1. $P(A) < 1$ for any event A.
2. $P(A^c) = 1 - P(A)$.
3. $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ for any two events A and B.

Proving Observations

Observation 1: $P(\emptyset) = 0$

$$\begin{aligned} P(\emptyset) &= 0 \\ \emptyset \cup S &= S \\ \emptyset \cap S &= \emptyset \\ P(\emptyset \cup S) &= P(S) \\ P(\emptyset) + P(S) &= P(S) \quad \text{(by axiom 3)} \end{aligned}$$

$\emptyset$ and S are mutually exclusive.
Therefore, $P(\emptyset) = 0$ by algebra.

Observation 2: $P(A^c) = 1 - P(A)$

$$\begin{aligned}
S &= A \cup A^c \\
\emptyset &= A \cap A^c \\
P(A^c \cup A) &= P(S) \\
P(A^c) + P(A) &= P(S) \quad \text{(by axiom 3)} \\
P(A^c) + P(A) &= 1 \\
P(A^c) &= 1 - P(A)
\end{aligned}$$

Observation 3: $P(A \cup B) = P(A) + P(B)$ **if** $A \cap B = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad \text{for mutually exclusive } A_i$$

Let $A_1 = A,\ A_2 = B,\ A_3 = A_4 = \emptyset$

$$\begin{aligned}
P(A \cup B \cup \emptyset \cup \emptyset) &= P(A) + P(B) + 0 \\
P(A \cup B) &= P(A) + P(B) \quad \text{(by axiom 3)}
\end{aligned}$$

Identities for the next proof

$$\begin{aligned}
P(A) &= P(A \cap B') + P(A \cap B) \\
P(B) &= P(A' \cap B) + P(A \cap B) \\
P(A \cup B) &= P(A \cap B') + P(A \cap B) + P(A' \cap B) \\
1 &= P(A \cap B') + P(A \cap B) + P(A' \cap B) + P(A' \cap B')
\end{aligned}$$

Observation 4: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\begin{aligned}
P(A \cup B) &= P(A \cap B') + P(A \cap B) + P(A' \cap B) \\
&= P(A) + P(A' \cap B) \\
&= P(A) + [P(B) - P(A \cap B)] \\
&= P(A) + P(B) - P(A \cap B)
\end{aligned}$$

Other Propositions

$A \subseteq B \implies P(A) \leq P(B)$

$$\begin{aligned}
B &= (A \cap B) \cup (A' \cap B) \\
A \cap B &= A \\
B &= A \cup (A' \cap B) \\
P(B) &= P(A) + P(A' \cap B) \quad \text{(by axiom 3)} \\
P(B) &\geq P(A) \quad \text{since } P(A' \cap B) \geq 0 \text{ (by axiom 1)}
\end{aligned}$$

For any A, $A \subseteq S$

$$\begin{aligned}
P(A) &\leq P(S) \\
\therefore P(A) &< 1
\end{aligned}$$

Example If $P(C) = .46, P(D) = .47, P(C \cup D) = .62$, find the following:

$$\begin{aligned}
P(C') &= 1 - P(C) = 1 - .46 = .54 \\
P(C \cap D) &= P(C) + P(D) - P(C \cup D) = .46 + .47 - .62 = .31 \\
P(C \cap D') &= P(C) - P(C \cap D) = .46 - .31 = .15 \\
P(C' \cup D') &= 1 - P(C \cup D) = 1 - .62 = .38
\end{aligned}$$

Probability Square Trick

Example There are 5 simple events in a sample space. $S = E_1, E_2, E_3, E_4, E_5$ with probabilities p_1, p_2, p_3, p_4, p_5. Suppose $p_3 = 2p_2 = 2p_4$ and $p_2 = 2p_1 = 2p_5$. Find p_1, p_2, p_3, p_4, p_5.

$$\begin{aligned}
p_1 + p_2 + p_3 + p_4 + p_5 &= 1 \\
p_3 &= 2p_2 \\
p_3 &= 2p_4 \\
p_4 &= p_2 \\
p_2 &= 2p_1 \\
p_2 &= 2p_5 \\
p_1 &= p_5
\end{aligned}$$

Solve for the 5 unknowns with the 5 equations:

2.3 Counting Techniques

Equally Likely Outcomes

If all outcomes in a sample space are equally likely for any event A, then

$$P(A) = \frac{\text{Number of outcomes in A}}{\text{Number of outcomes in S}} = \frac{n(A)}{n(S)}$$

where $n(A)$ is the number of outcomes in event A and $n(S)$ is the number of outcomes in sample space S.

Combinatorics

Combinatorics is the branch of mathematics dealing with combinations of objects in specific sets under certain constraints. It provides tools for counting and arranging objects. Some basic combinatorial concepts include:

- **Permutations:** The number of ways to arrange a set of objects in a specific order.
- **Combinations:** The number of ways to select a subset of objects from a larger set, without regard to the order of selection.

Counting Subsets Let $\Omega = a, b, c, d, e, f, g, h, i, j$. How would we find all possible subsets of Ω?

$$Num = 2^{len(\Omega)} = 2^{10} = 1024$$

This is because each element can either be included or excluded (2 options) from a subset, leading to 2^n possible subsets for a set with n elements.

Permutations and Combinations

Permutations A permutation is an arrangement of objects in a specific order. The number of permutations of k objects chosen from n distinct objects is given by the following formula:

$$P_{k,n} = \frac{n!}{(n-k)!} = n \times (n-1) \times (n-2) \times ... \times (n-k+1)$$

Where $P_{k,n}$ is the number of permutations of n objects taken k at a time, $n!$ is the factorial of n, and $(n-k)!$ is the factorial of $(n-k)$.

Example: Suppose we want to rank 5 candidates Amy, Bai, Catherine, David, and Eduardo in preference for a position. How many different ways can we rank (1.) all of the candidates and (2.) only 2 of the candidates?

$$1.\quad P_{5,5} = \frac{5!}{5-5!} = \frac{5!}{1} = 120$$
$$2.\quad P_{5,2} = \frac{5!}{5-2!} = \frac{5!}{3!} = 20$$

Combinations A combination is a selection of objects without regard to the order of selection. The number of combinations of k objects chosen from n distinct objects is given by the following formula:

$$C_{k,n} = \frac{n!}{k!(n-k)!} = \frac{P_{k,n}}{k!}$$

Where $C_{k,n}$ is the number of combinations of n objects taken k at a time, $n!$ is the factorial of n, $k!$ is the factorial of k, and $(n-k)!$ is the factorial of $(n-k)$.

Example: How many hands of 3 cards can be drawn from a standard deck of 52 playing cards?

$$C_{3,52} = \binom{52}{3} = \frac{52!}{3!(52-3)!} = \frac{52!}{3!49!} = 22100$$

Example: A playlist contains 100 songs. 10 of these songs are by The Strokes. We can find the probability of the first song by The Strokes being the 5th song played by finding the ratio of the number of favorable outcomes to the total number of outcomes.

$$\begin{aligned}
P(\text{1st Strokes on 5th}) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
&= \frac{P_{4,90} \cdot P_{1,10}}{P_{5,100}} = \frac{\frac{90!}{(90-4)!} \cdot 10}{\frac{100!}{(100-5)!}} \\
&= .0679
\end{aligned}$$

Another way of thinking about this would be that there are $\binom{100}{10}$ different ways to choose the 10 Strokes songs. Now if we choose 9 of the last 95 songs to be Strokes songs, there are $\binom{95}{9}$ ways to do this, leaving 4 non-Strokes songs and 1 Strokes song in the first 5 songs. The probability is then:

$$P(\text{1st Strokes on 5th}) = \frac{\binom{95}{9}}{\binom{100}{10}} = .0679$$

Example: A university warehouse has recieved 25 printers, of which 10 are laser printers and 15 are inkjet printers. If 6 printers are randomly selected for testing, what is the probability that exacty 3 of those selected are laser printers?

Let D_3 be the event that exactly 3 of the selected printers are laser printers.

$$\begin{aligned}
P(D_3) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
&= \frac{N(D_3)}{N} = \frac{\binom{15}{3} \cdot \binom{10}{3}}{\binom{25}{6}} = .3083
\end{aligned}$$

2.4 Conditional Probability

A conditional probability of an event is the probability that the event occurs given that some other event has already occurred. Conditional probabilities can prove to be counterintuitive and may lead tableofcontents suprising results.

Definition: The conditional probability of an event A given that an event B has occurred is denoted by $P(A|B)$ and is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: If you flip 3 coins, what is the probability that exactly 2 are heads (B)?

$$\begin{aligned}
&\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} \\
&B = \{HHT, HTH, THH\} \\
&A = \{HHH, HTT, HTH, HHT\}
\end{aligned}$$

$$\begin{aligned}
P(A \cap B) &= \frac{2}{8} \\
P(B) &= \frac{3}{8} \\
P(A|B) = \frac{P(A \cap B)}{P(B)} &= \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{2}{3}
\end{aligned}$$

Multiplication Rule

The multiplication rule is a way to find the probability of the intersection of two events. It states that:

$$\begin{aligned}
P(A \cap B) = P(A|B) \cdot P(B) \\
P(A \cap B) = P(B|A) \cdot P(A)
\end{aligned}$$

CHECK THIS FOR ACCURACY

General Multiplication Rule: For any number of events $A_1, A_2, \ldots, A_n$:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1)\cdot P(A_2|A_1)\cdot P(A_3|A_1\cap A_2)\cdots P(A_n|A_1\cap A_2\cap\cdots\cap A_{n-1})$$

Law of Total Probability

For any events A and B, we can get the following from the multiplication rule:

$$P(A\cap B) = P(A|B)P(B)$$
$$P(A\cap B') = P(A|B')P(B')$$

: Note that B and B' form a *partition* of Ω.

General Law of Total Probability: Let $B_1, \ldots, B_n$ be a partition of the sample space Ω. Then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

Example: Suppose that we have 2 boxes with marbles in them. Box 1 has 1 white and 3 black marbles. Box 2 has 2 white and 2 black marbles. We randomly select a box and then randomly select a marble from that box. What is the probability that we select a black marble?

$$\begin{aligned} P(Black) &= P(Black|Box1)P(Box1) + P(Black|Box2)P(Box2) \\ &= \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{2}{4}\right)\left(\frac{1}{2}\right) \\ &= \frac{3}{8} + \frac{2}{8} = \frac{5}{8} \end{aligned}$$

Example: On any given morning, it is either raining (25% of the time), foggy (35% of the time), or sunny (40% of the time). If it is raining, there is a 30% change that I will miss the bus. If it is foggy, there is a 20% chance that I will miss the bus. If it is sunny, there is a 10% chance that I will miss the bus. What is the probability that I will miss the bus on any given morning?

$$\begin{aligned} P(MB) &= P(MB|R)P(R) + P(MB|F)P(F) + P(MB|S)P(S) \\ &= (0.30)(0.25) + (0.20)(0.35) + (0.10)(0.40) \\ &= 0.075 + 0.07 + 0.04 = 0.185 \end{aligned}$$

Bayesian Analysis

Bayesian analysis is a statistical method that involves using Bayes' theorem to update the probability of a hypothesis based on new evidence or data. It is named after Thomas Bayes, an 18th-century statistician and theologian.

Bayes' Theorem Let $B_1, \ldots, B_n$ form a partition of the sample space Ω. Then for any event A:

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{n} P(A|B_j)P(B_j)}$$

Note that in the special case of the partition consisting of B and B', this reduces to:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

Example: If I pull a black marble from one of the boxes in the previous example, what is the probability that I pulled it from Box 1?

$$\begin{aligned} P(B_1|Black) &= \frac{P(Black|B_1)P(B_1)}{P(Black)} \\ &= \frac{(\frac{3}{4})(\frac{1}{2})}{\frac{5}{8}} = \frac{3}{5} \end{aligned}$$

Example: According to the Arizona Chapter of the American Lung Association, 7% of the population has lung disease. Of those, people, 90% are smokers; and of those without lung disease, 74.7% are non-smokers. What are the chances that a smoker has lung disease?

$$\begin{aligned} P(LD|S) &= \frac{P(S|LD)P(LD)}{P(S|LD)P(LD) + P(S|LD')P(LD')} \\ &= \frac{(0.90)(0.07)}{(0.90)(0.07) + (0.253)(0.93)} \\ &= \frac{0.063}{0.063 + 0.23529} = \frac{0.063}{0.29829} \approx 0.2111 \end{aligned}$$

Example: You give a friend a letter to mail. He forgets to mail it with probability 0.2. Given that he mails it, the Post Office delivers it with probability 0.9. Given that the letter was not delivered, what is the probability that it was not mailed.

$$\begin{aligned} P(M'|D') &= \frac{P(D'|M')P(M')}{P(D'|M')P(M') + P(D'|M)P(M)} \\ &= \frac{(1)(0.2)}{(1)(0.2) + (0.1)(0.8)} \\ &= \frac{0.2}{0.2 + 0.08} = \frac{0.2}{0.28} = \frac{5}{7} \end{aligned}$$